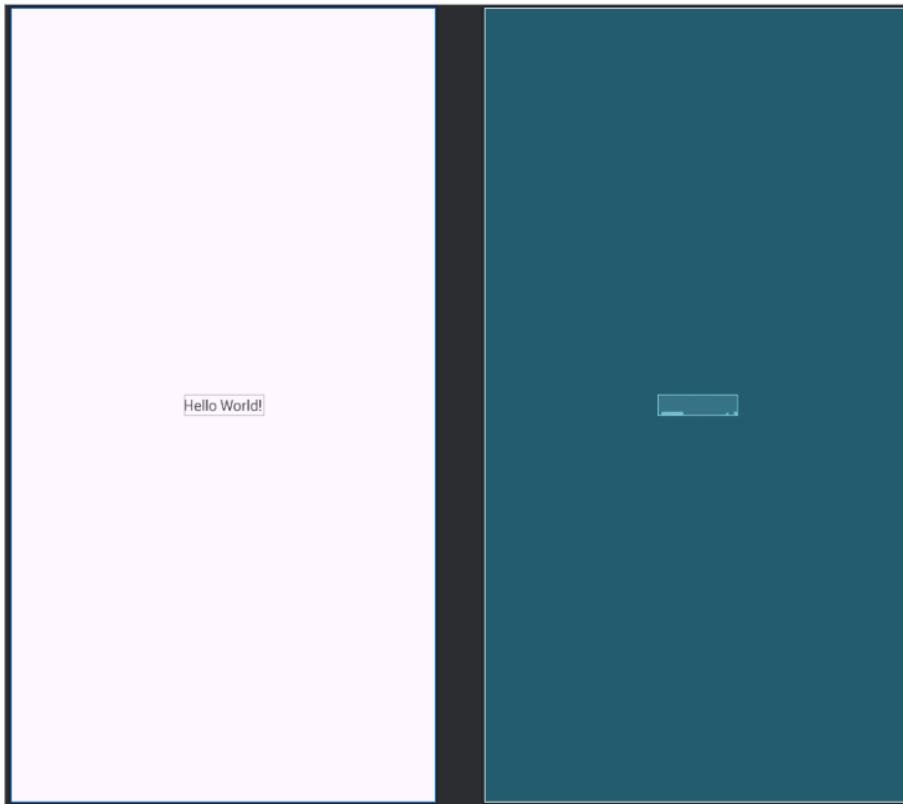


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Program 01: Creating “Hello World” Application.

Design:



Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:text=" Hello World " />

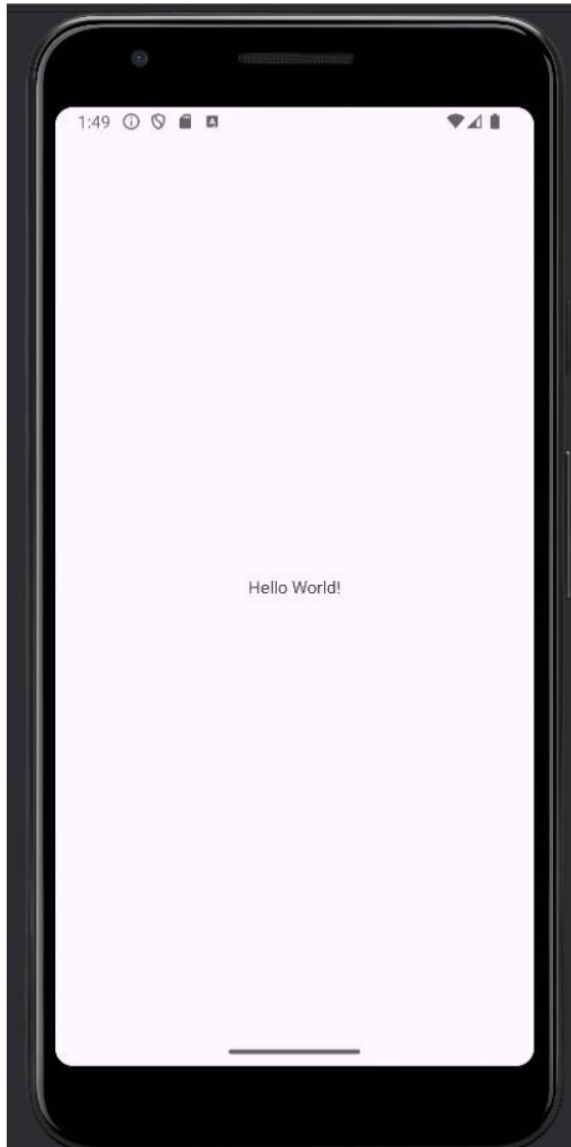
</RelativeLayout>
```

MainActivity.java

```
package com.helloworld;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity1 extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
    }
}
```

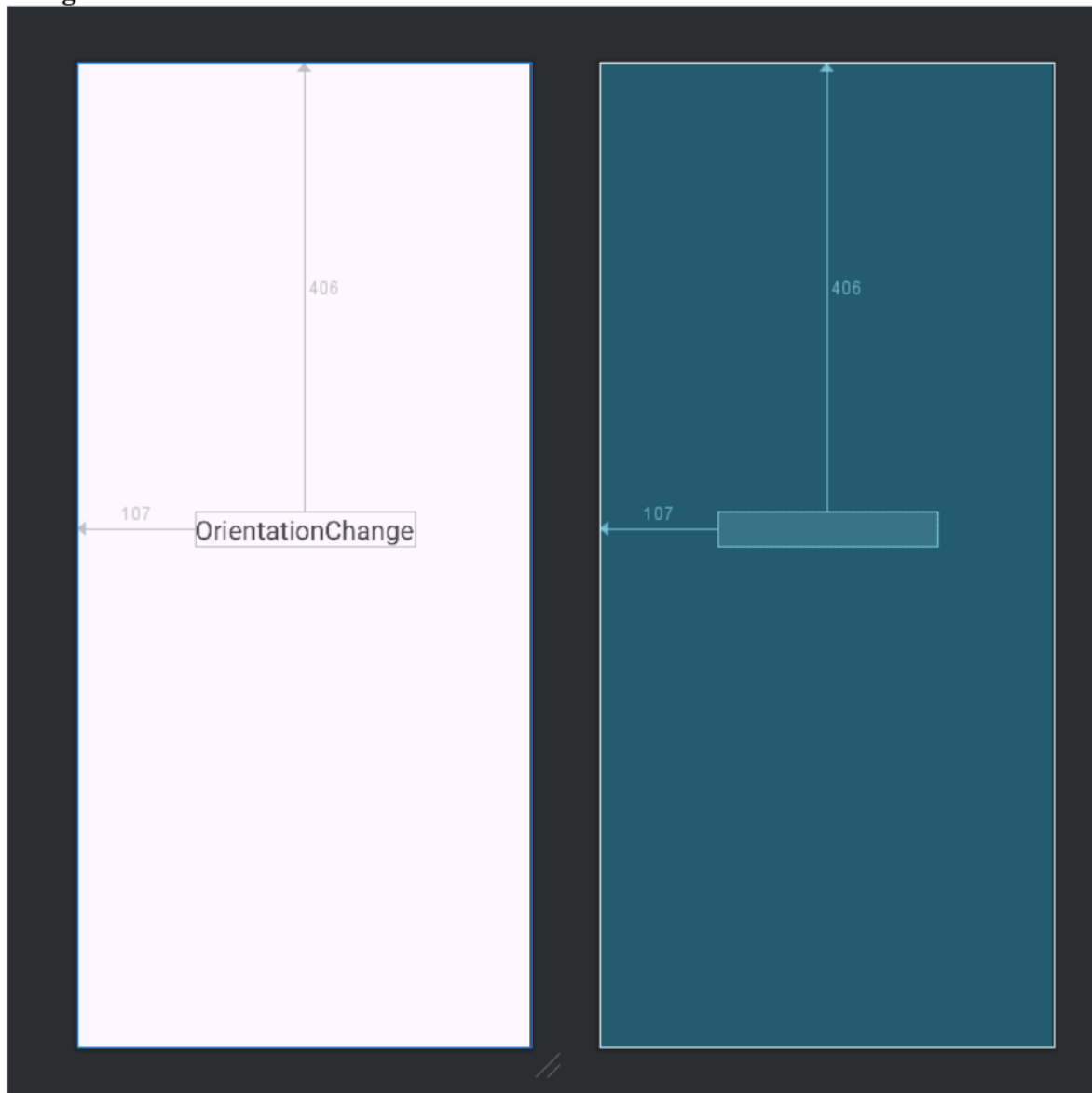
AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Helloworld"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Output:

Program 02: Creating an application that displays message based on the screen orientation

Design:



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```

<TextView
    android:id="@+id/orientationchange"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:layout_marginStart="107dp"
    android:layout_marginTop="406dp"
    android:text="OrientationChange"
    android:textSize="24sp" />
</RelativeLayout>

```

MainActivity.java

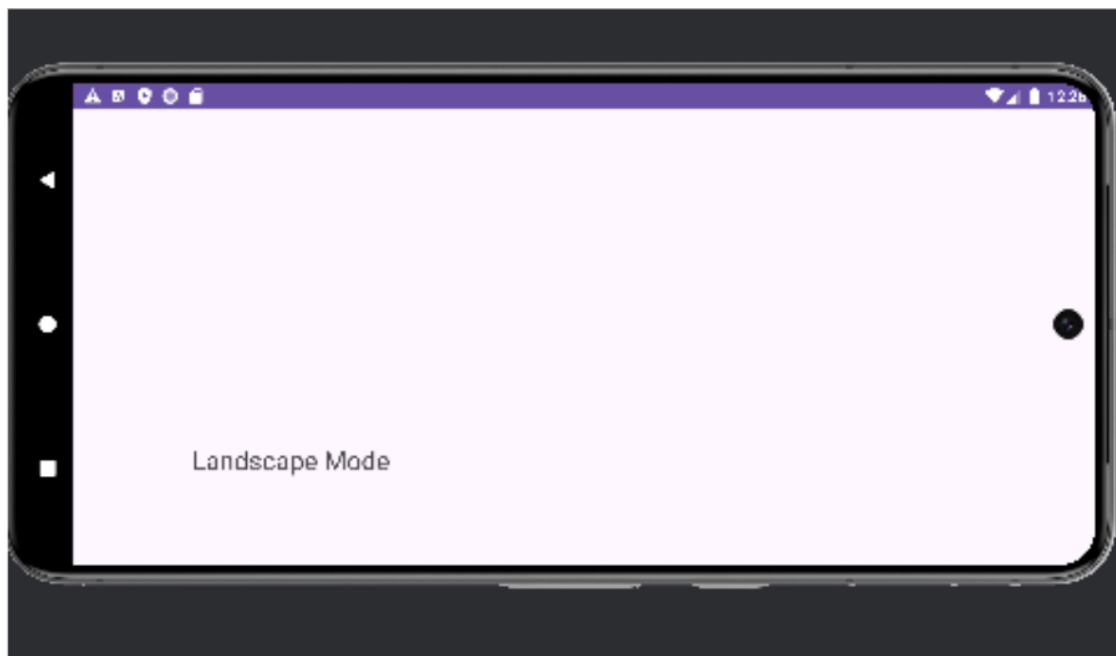
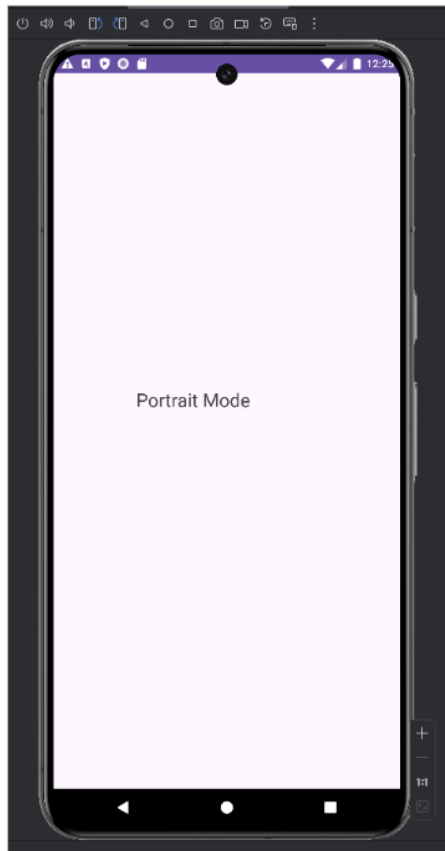
```

package com.example.orientationchange;
import android.graphics.Point;
import android.os.Bundle;
import android.view.Display;
import android.view.WindowManager;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        TextView orientationMessage=findViewById(R.id.orientationchange);
        WindowManager wm=(WindowManager) getSystemService(WINDOW_SERVICE);
        Display d=wm.getDefaultDisplay();
        Point size=new Point();
        d.getSize(size);
        int w=size.x;
        int h=size.y;
        if (w>h) {
            orientationMessage.setText("Landscape Mode");
        }
        else {
            orientationMessage.setText("Portrait Mode");
        }
    }
}

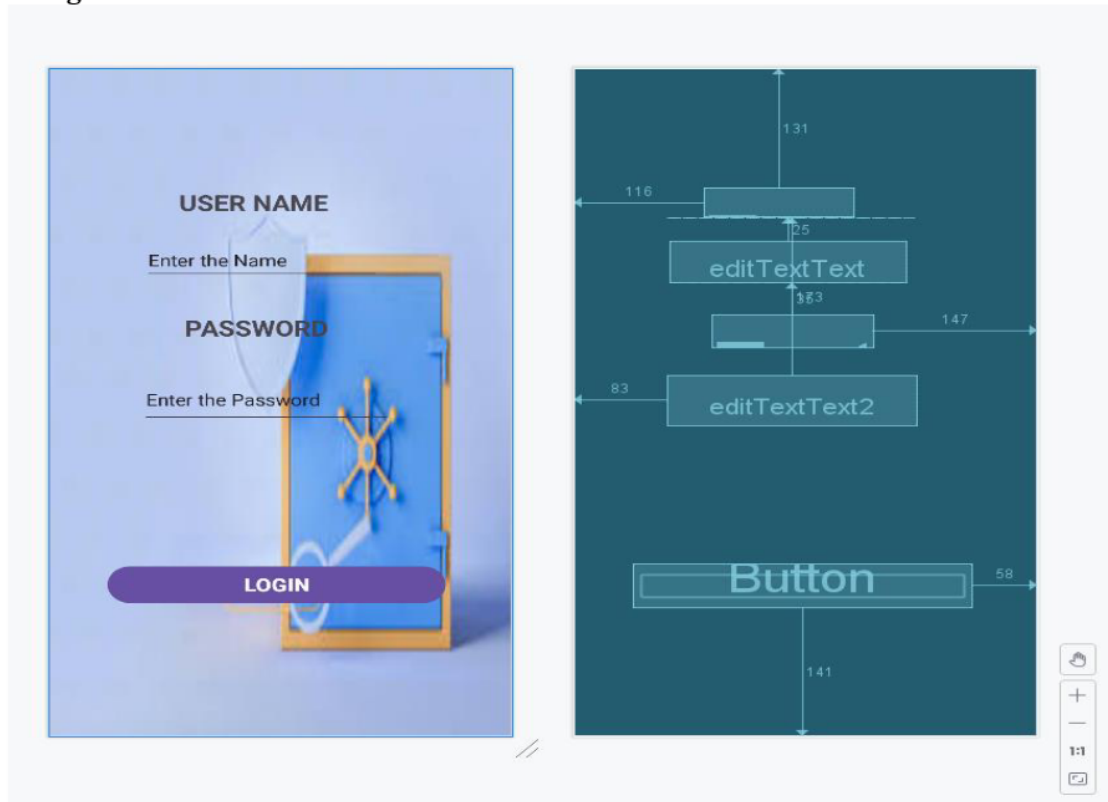
```

OUTPUT



Program 03: Developing a Login Window Using UI Controls

Design:



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/images"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_marginStart="116dp"
        android:layout_marginTop="131dp"
        android:text="USER NAME"
        android:textSize="24sp"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/textView"
        android:layout_marginStart="85dp"
```



```

        android:layout_marginTop="25dp"
        android:ems="10"
        android:inputType="text"
        android:text="Enter the Name" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="143dp"
    android:layout_height="36dp"
    android:layout_below="@id/editTextText"
    android:layout_alignParentEnd="true"
    android:layout_marginTop="35dp"
    android:layout_marginEnd="147dp"
    android:text="PASSWORD"
    android:textSize="24sp"
    android:textStyle="bold" />

<EditText
    android:id="@+id/editTextText2"

    android:layout_width="222dp"
    android:layout_height="55dp"
    android:layout_below="@id/textView"
    android:layout_alignParentStart="true"
    android:layout_marginStart="83dp"
    android:layout_marginTop="173dp"
    android:ems="10"
    android:inputType="text"
    android:text="Enter the Password" />

<Button
    android:id="@+id/button"
    android:layout_width="301dp"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="58dp"
    android:layout_marginBottom="141dp"
    android:text="LOGIN"
    android:textSize="20sp"
    android:textStyle="bold" />
</RelativeLayout>

```

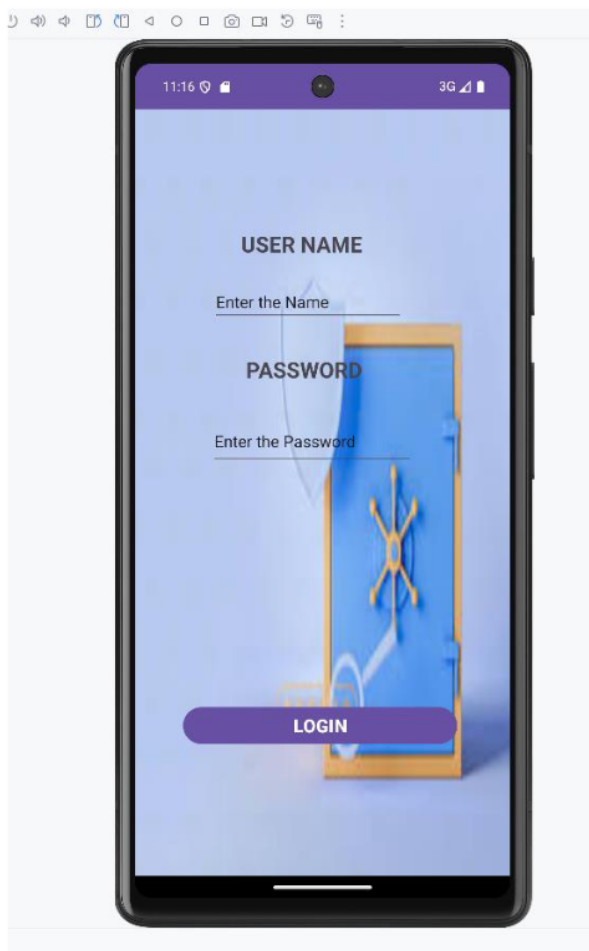
MainActivity.java

```

package com.example.loginapp;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

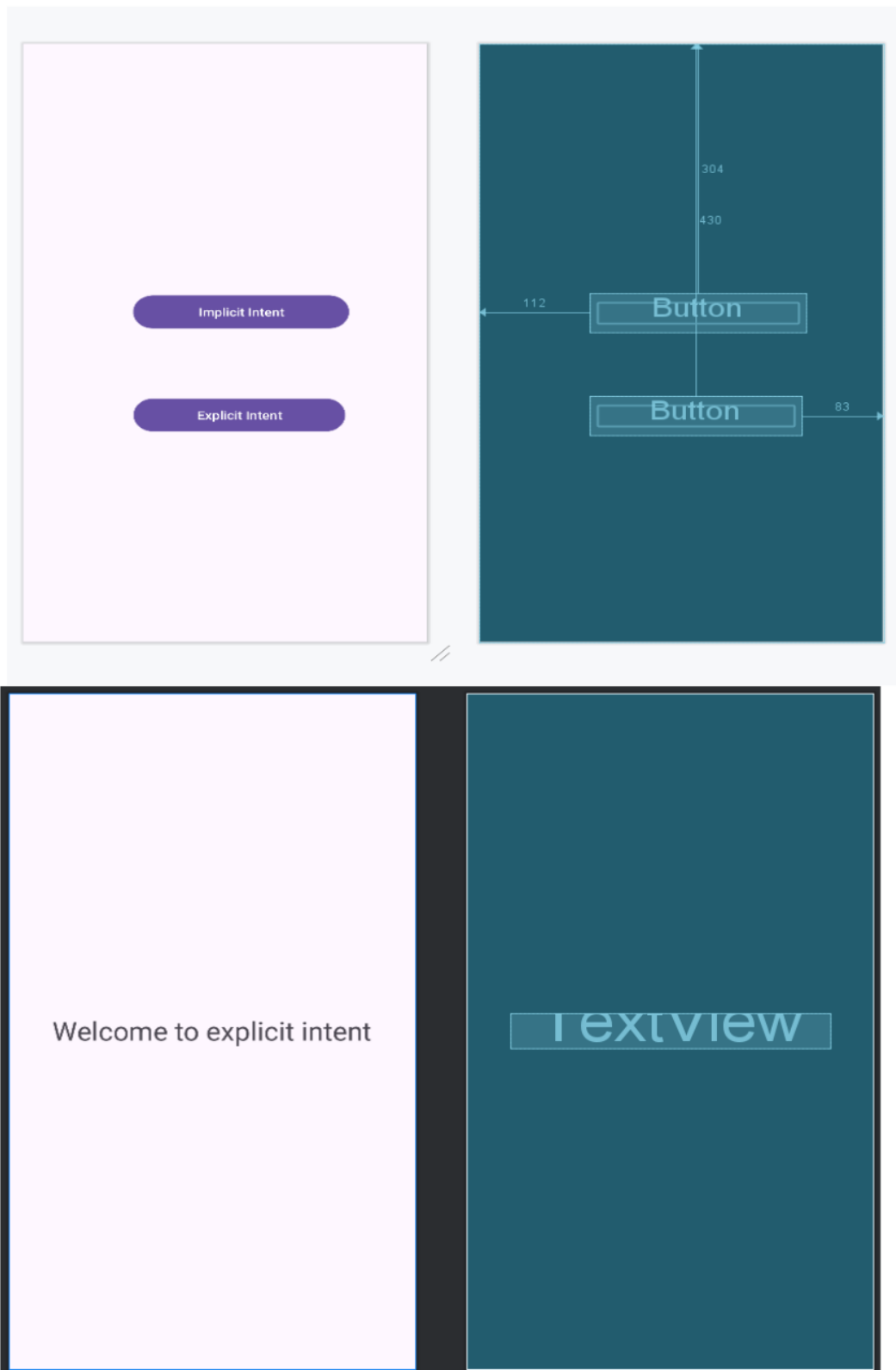
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

OUTPUT

Program 04: Create an application to implement new activity using explicit intent and implicit intent.

Design



Code:**activity_main.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="220dp"
        android:layout_height="wrap_content"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_marginStart="112dp"
        android:layout_marginTop="304dp"
        android:onClick="onImplicitButtonClicked"
        android:text="Implicit Intent"
        tools:layout_editor_absoluteX="36dp"
        tools:layout_editor_absoluteY="286dp" />

    <Button
        android:id="@+id/button2"
        android:layout_width="216dp"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_alignParentEnd="true"
        android:layout_marginTop="430dp"
        android:layout_marginEnd="83dp"
        android:onClick="onExplicitButtonClicked"
        android:text="Explicit Intent"
        tools:layout_editor_absoluteX="248dp"
        tools:layout_editor_absoluteY="286dp" />

</RelativeLayout>

```

activity_new.xml

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <TextView
        android:id="@+id/tv"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:text="Welcome to explicit intent"/>

</RelativeLayout>

```

MainActivity.java

```

package com.example.l4;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void onImplicitButtonClicked(View view) {
        Uri url = Uri.parse("https://www.google.com");
        Intent i = new Intent(Intent.ACTION_VIEW, url);
        startActivity(i);
    }
    public void onExplicitButtonClicked(View view) {
        Intent i=new Intent(MainActivity.this,MainActivity2.class);
        startActivity(i);
    }
}

```

NewActivity.java

```

package com.example.l4;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
    }
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.L4"
        tools:targetApi="31">

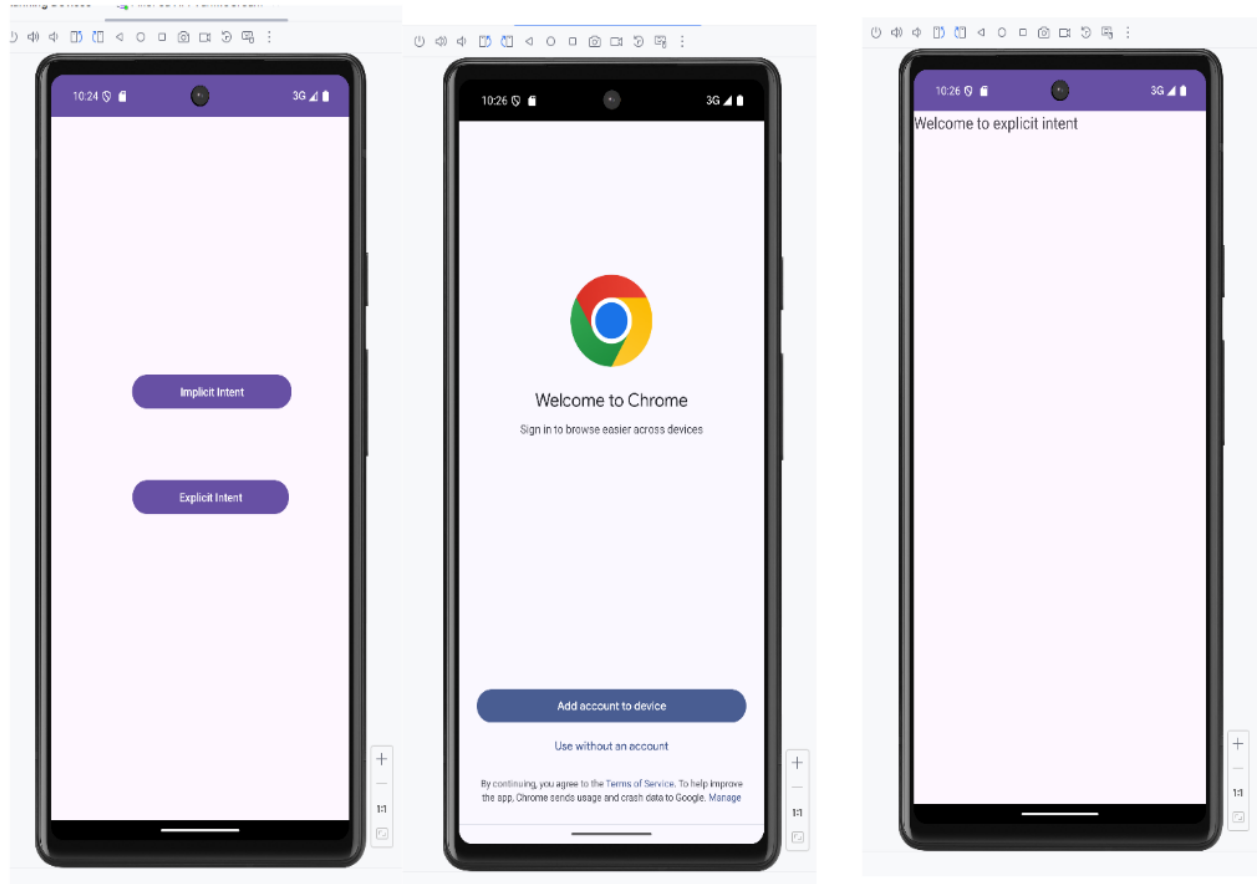
```

```

<activity
    android:name=".MainActivity2"
    android:exported="false" />
<activity
    android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

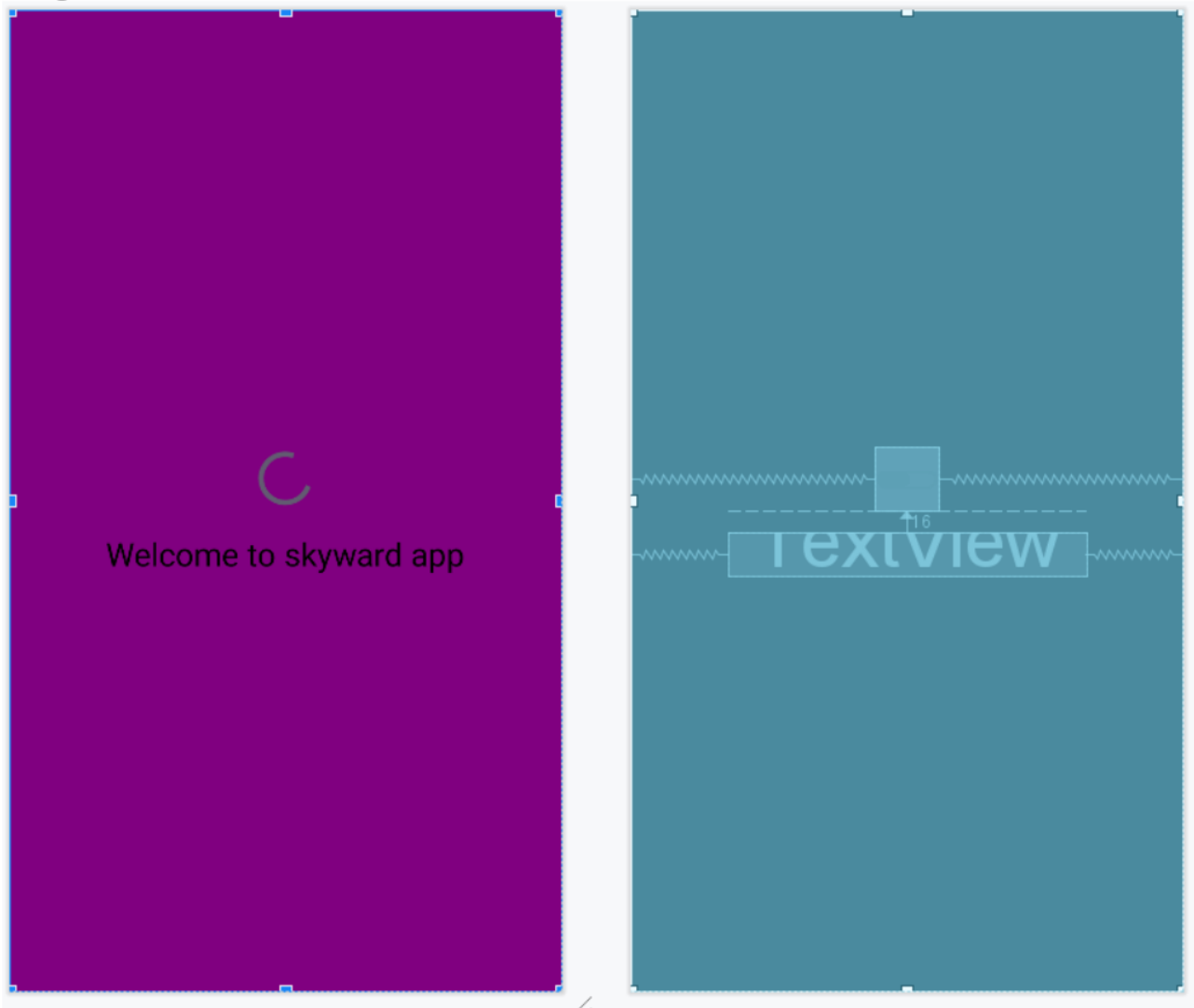
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>
</manifest>

```



Program 05: Create an application that displays custom designed Opening Screen

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <RelativeLayout
        android:id="@+id/splash_screen"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:background="#800080"
        android:gravity="center"
        android:padding="16dp">

        <ProgressBar
            android:id="@+id/progress_Bar"
            style="?android:attr/progressBarStyle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="16dp"
            android:layout_marginTop="16dp"
            android:layout_centerHorizontal="true"
```

```

        android:indeterminate="true"/>

<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Welcome to skyward app"
    android:textSize="24dp"
    android:textColor="#FF000000"
    android:layout_below="@+id/progress_Bar"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="16dp"/>
</RelativeLayout>

<LinearLayout
    android:id="@+id/main_content"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center"
    android:visibility="gone">

    <TextView
        android:id="@+id/welcome_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="welcome to the main activity"
        android:textSize="18sp"
        android:textColor="#0000FF"/>
</LinearLayout>
</FrameLayout>

```

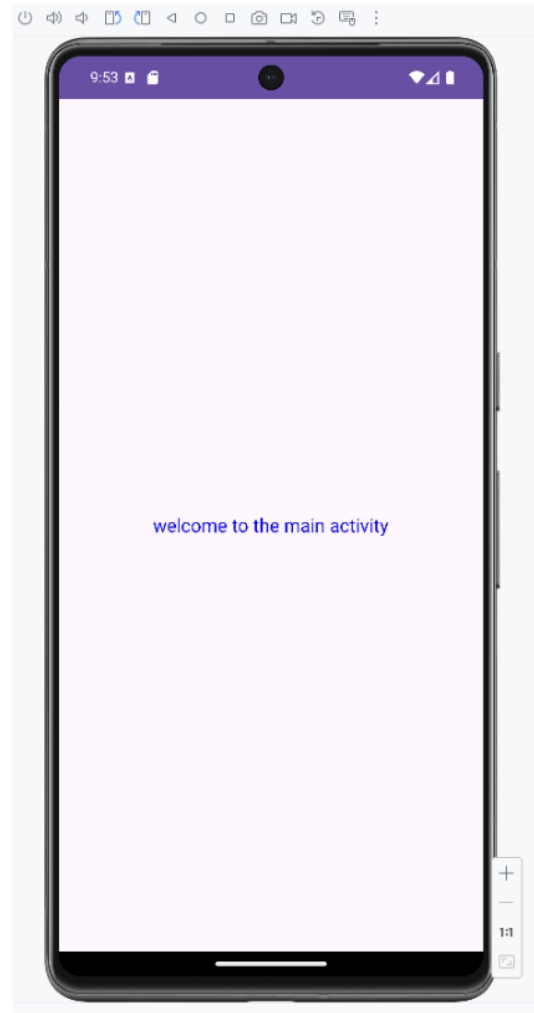
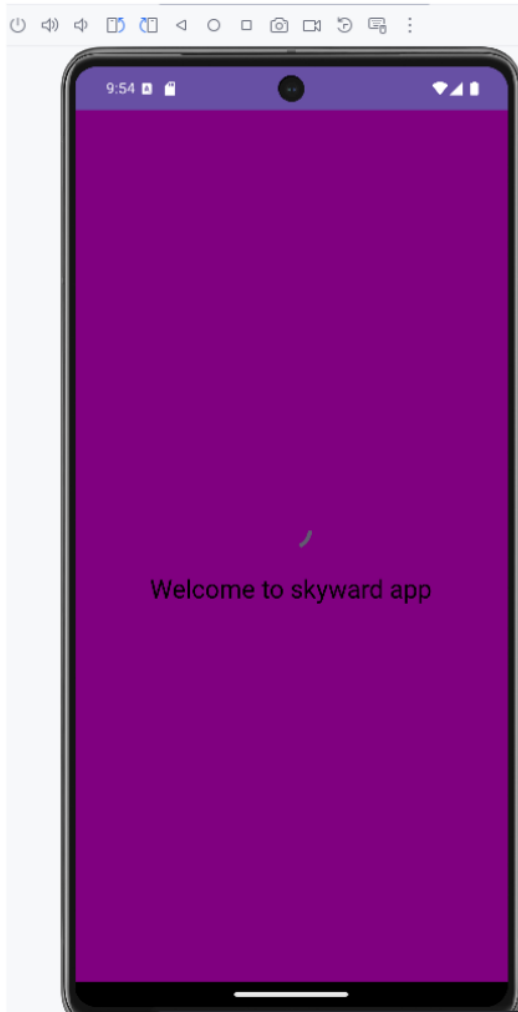
MainActivity.java

```

package com.example.splashscreen;
import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    private static final int SPLASH_DISPLAY_LENGTH=10000;

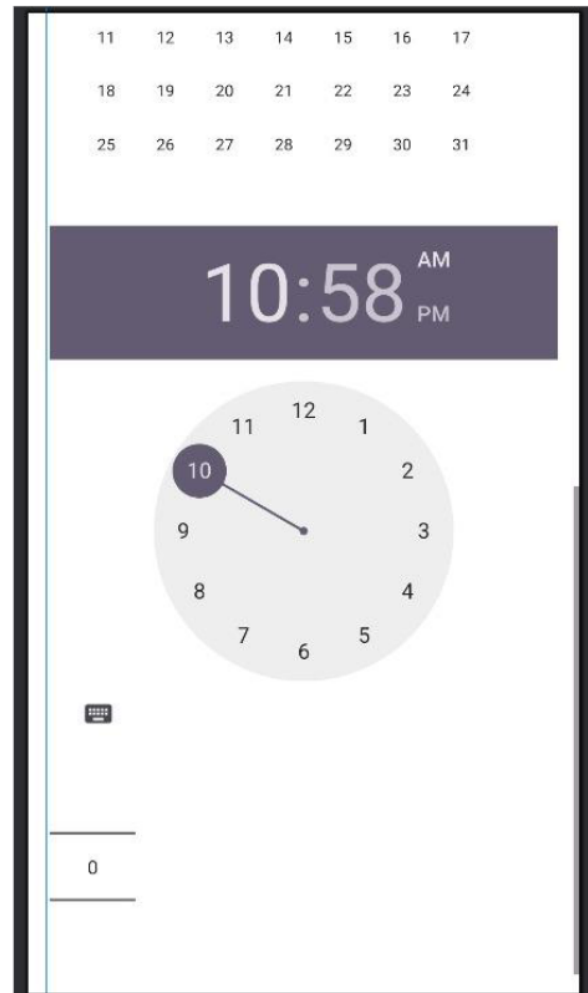
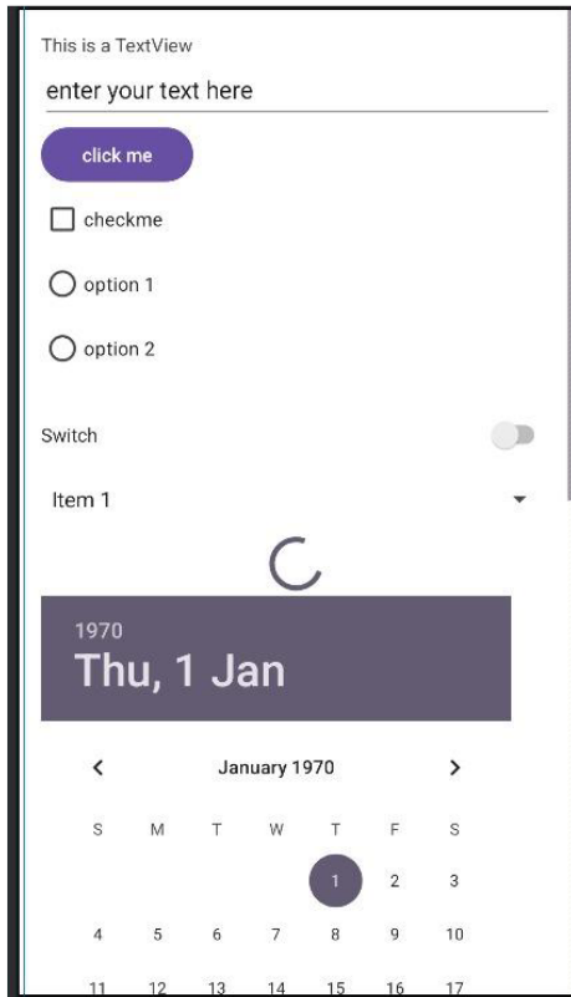
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final View splashscreen=findViewById(R.id.splash_screen);
        final View mainContent=findViewById(R.id.main_content);
        new Handler().postDelayed(new Runnable()
        {
            @Override
            public void run()
            {
                splashscreen.setVisibility(View.GONE);
                mainContent.setVisibility(View.VISIBLE);
            }
        },SPLASH_DISPLAY_LENGTH);
    }
}

```


OUTPUT

Program 06: Create an UI with All Views.

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/white"
        android:orientation="vertical"
        android:padding="16dp">
```

```
<TextView
    android:id="@+id/textView"
    android:layout_width="243dp"
    android:layout_height="wrap_content"
    android:text="This is a TextView" />

<EditText
    android:id="@+id/editText"
    android:layout_width="381dp"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="text"
    android:minHeight="48dp"
    android:text="enter your text here" />

<Button
    android:id="@+id/button"
    android:layout_width="113dp"
    android:layout_height="wrap_content"
    android:text="click me" />

<CheckBox
    android:id="@+id/checkBox"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="checkme" />

<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:layout_marginBottom="16dp">

    <RadioButton
        android:id="@+id/radioButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="option 1" />

    <RadioButton
        android:id="@+id/radioButton3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="option 2" />
</RadioGroup>
```

```
<Switch
    android:id="@+id/switch1"
    android:layout_width="378dp"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    android:text="Switch" />

<Spinner
    android:id="@+id/spinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp" />

<ProgressBar
    android:id="@+id/progressBar"
    style="?android:attr/progressBarStyle"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

<DatePicker
    android:id="@+id/datepicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<TimePicker
    android:id="@+id/timepicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<NumberPicker
    android:id="@+id/numberpicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginBottom="16dp"/>

</LinearLayout>
</ScrollView>
```

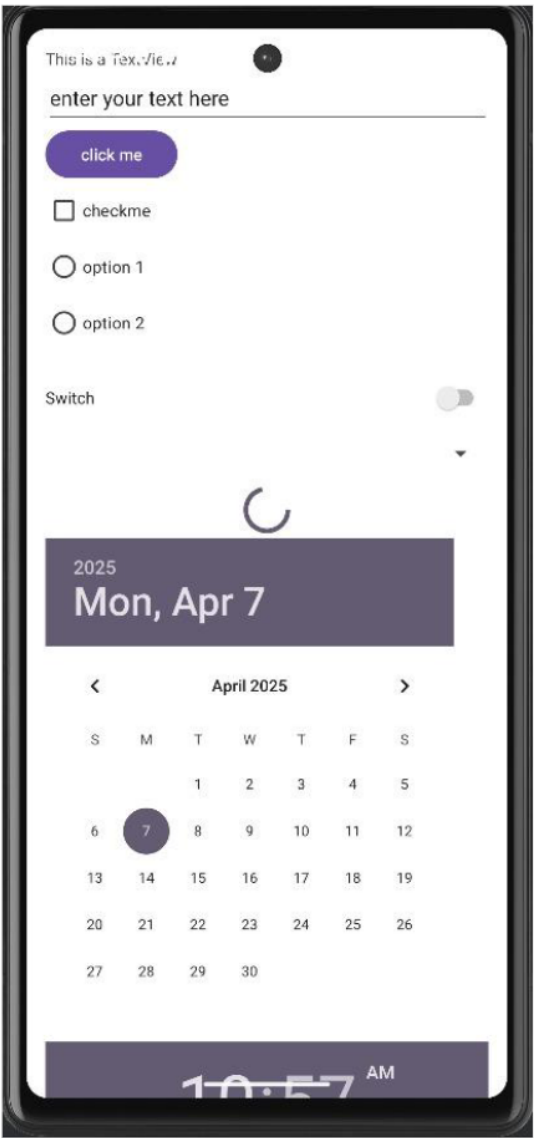
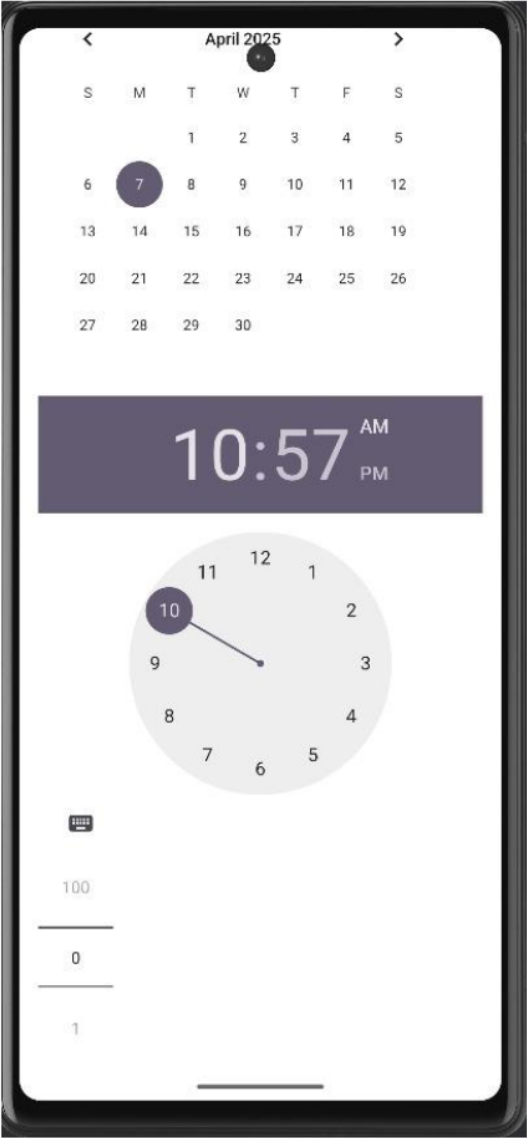
MainActivity.java

```
package com.example.timedate;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.DatePicker;
import android.widget.EditText;
import android.widget.NumberPicker;
import android.widget.ProgressBar;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Spinner;
```

```
import android.widget.Switch;
import android.widget.TextView;
import android.widget.TimePicker;
import androidx.appcompat.app.AppCompatActivity;
import com.example.timedate.R;

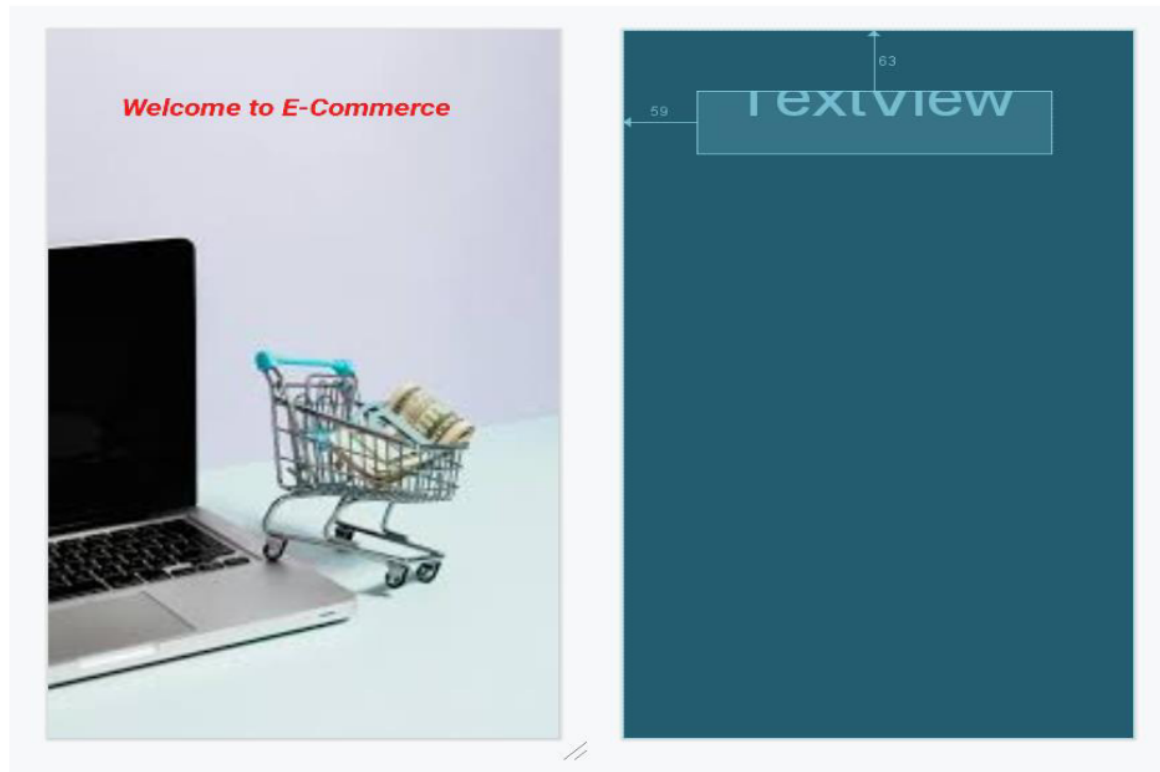
public class MainActivity extends AppCompatActivity {
    private TextView textView;
    private EditText editText;
    private Button button;private CheckBox checkBox;
    private RadioGroup radioGroup;
    private RadioButton radioButton1;
    private RadioButton radioButton2;
    private Switch switch1;
    private Spinner spinner;
    private ProgressBar progressBar;
    private DatePicker datePicker;
    private TimePicker timePicker;
    private NumberPicker numberPicker;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        textView = findViewById(R.id.textView);
        editText = findViewById(R.id.editText);
        button = findViewById(R.id.button);
        checkBox = findViewById(R.id.checkBox);
        radioGroup=findViewById(R.id.radioGroup);
        radioButton1 = findViewById(R.id.radioButton);
        radioButton2 =findViewById(R.id.radioButton3);
        switch1=findViewById(R.id.switch1);
        spinner =findViewById(R.id.spinner);
        progressBar = findViewById(R.id.progressBar);
        datePicker =findViewById(R.id.datepicker);
        timePicker = findViewById(R.id.timepicker);
        numberPicker =findViewById(R.id.numberpicker);
        numberPicker.setMinValue(0);
        numberPicker.setMaxValue (100);
    }
}
```

OUTPUT

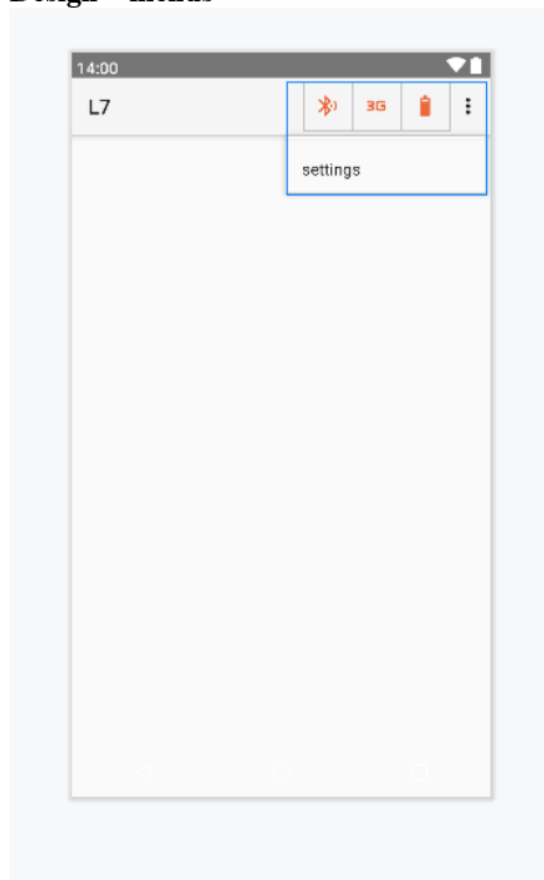


Program 07: Create menu in Application

Design - activity_main



Design – menus



activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

    android:background="@drawable/e_commerce"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="286dp"
        android:layout_height="65dp"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_marginStart="59dp"
        android:layout_marginTop="63dp"
        android:text="Welcome to E-Commerce"
        android:textColor="#F11D1D"
        android:textColorLink="#F31818"
        android:textSize="24sp"
        android:textStyle="bold|italic" />

</RelativeLayout>

```

menus.xml

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item
        android:id="@+id/search"
        android:icon="@drawable/search"
        app:showAsAction="always"
        android:title="search" />
    <item android:id="@+id/Refresh"
        android:title="Refresh"
        android:icon="@drawable/refresh"
        app:showAsAction="always" />
    <item android:id="@+id/Addtocart"
        android:title="Addtocart"/>
    <item android:id="@+id/settings"
        android:title="settings"/>
</menu>

```

MainActivity.java

```

package com.example.l7;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

```



```

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    @Override
    public boolean onCreatePanelMenu(int featureId, @NonNull Menu menu) {
        MenuInflater inflater = getMenuInflater();
        inflater.inflate(R.menu.menus, menu);
        return true;
    }
    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        if (item.getItemId() == R.id.search) {
            Toast.makeText(this, "Search Successfull", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.Refresh) {
            Toast.makeText(this, "Refresh Successfull", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.Addtocart) {
            Toast.makeText(this, "Addtocart Successfull", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.settings) {
            Toast.makeText(this, "Settings Clicked", Toast.LENGTH_SHORT).show();
        }
        return super.onOptionsItemSelected(item);
    }
}

```

AndroidManifest.xml

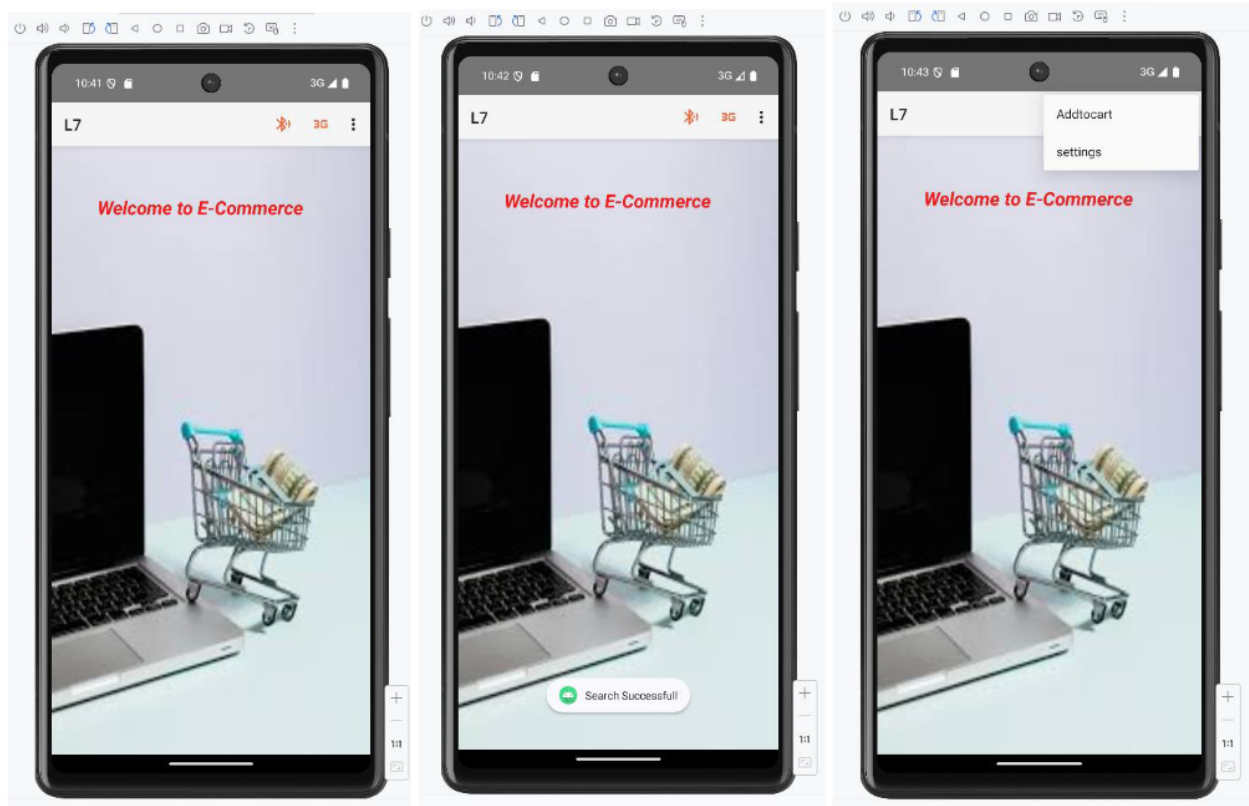
```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.AppCompat.Light"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

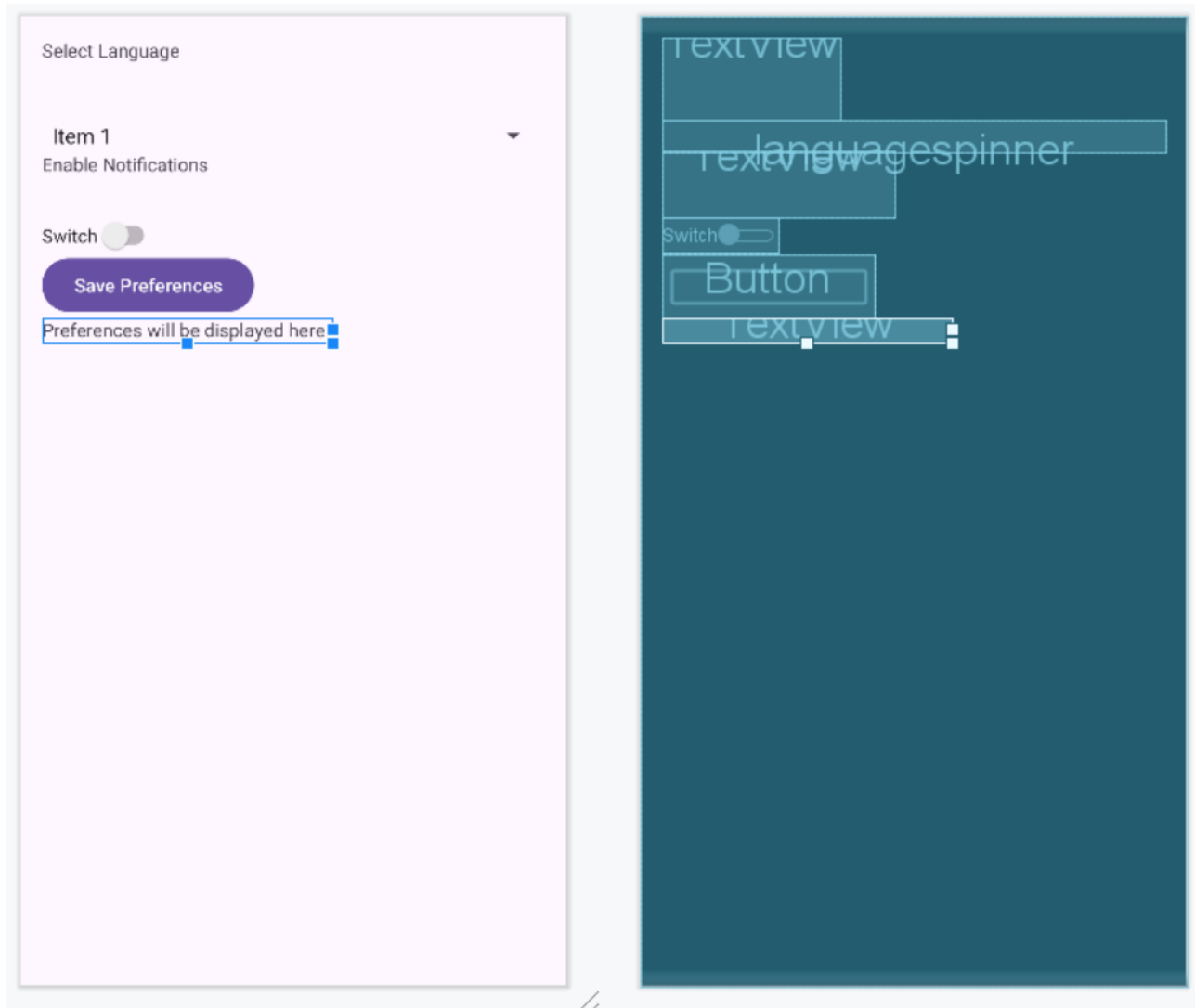
</manifest>

```

OUTPUT

Program 08: Read/Write the local data.

Design –



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="134dp"
        android:layout_height="62dp"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:text="Select Language" />
```

```

<Spinner
    android:id="@+id/languagespinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_alignParentTop="true"
    android:layout_alignParentEnd="true" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="175dp"
    android:layout_height="50dp"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Enable Notifications" />

<Switch
    android:id="@+id/notificationswitch"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Switch" />

<Button
    android:id="@+id/savebutton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Save Preferences" />

<TextView
    android:id="@+id/preferenceTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Preferences will be displayed here." />
</LinearLayout>

```

String.xml

```

<resources>
    <string name="app_name">userprefrence</string>
    <string-array name="languages">
        <item>English</item>
        <item>Spanish</item>
        <item>French</item>
        <item>german</item>
        <item>Chinese</item>
    </string-array>
</resources>

```

MainActivity.java

```

package com.example.userprefrence;
import android.content.Context;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.Spinner;
import android.widget.Switch;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity
{
    private Spinner languageSpinner;
    private Switch notificationsSwitch;
    private TextView preferencesTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        languageSpinner = findViewById(R.id.languagespinner);
        notificationsSwitch = findViewById(R.id.notificationswitch);
        preferencesTextView = findViewById(R.id.preferenceTextView);
        Button saveButton = findViewById(R.id.savebutton);
        loadUserPreferences();
        saveButton.setOnClickListener(v -> saveUserPreferences());
    }
    private void saveUserPreferences()
    {
        String language = languageSpinner.getSelectedItem().toString();
        boolean notificationsEnabled = notificationsSwitch.isChecked();
        SharedPreferences sharedPreferences = getSharedPreferences("UserPreferences",
Context.MODE_PRIVATE);
        SharedPreferences.Editor editor = sharedPreferences.edit();
        editor.putString("language", language);
        editor.putBoolean("notificationsEnabled", notificationsEnabled);
        editor.apply();
        preferencesTextView.setText(String.format("Language:%s\nNotificationsEnabled:%b", language,
notificationsEnabled));
    }
    private void loadUserPreferences()
    {
        SharedPreferences
sharedPreferences=getSharedPreferences("UserPreferences",Context.MODE_PRIVATE);
        String language=sharedPreferences.getString("language","English");
        boolean notificationsEnabled=sharedPreferences.getBoolean("notificationsEnabled",false);
        ArrayAdapter<CharSequence> adapter=ArrayAdapter.createFromResource(this,R.array.languages,
android.R.layout.simple_spinner_item);

```

```

adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
languageSpinner.setAdapter(adapter);
int spinnerPosition=adapter.getPosition(language);
languageSpinner.setSelection(spinnerPosition);
notificationsSwitch.setChecked(notificationsEnabled);

```

```

preferencesTextView.setText(String.format("Language:%s\nNotificationsEnabled:%b",language,notificationsEnabled));

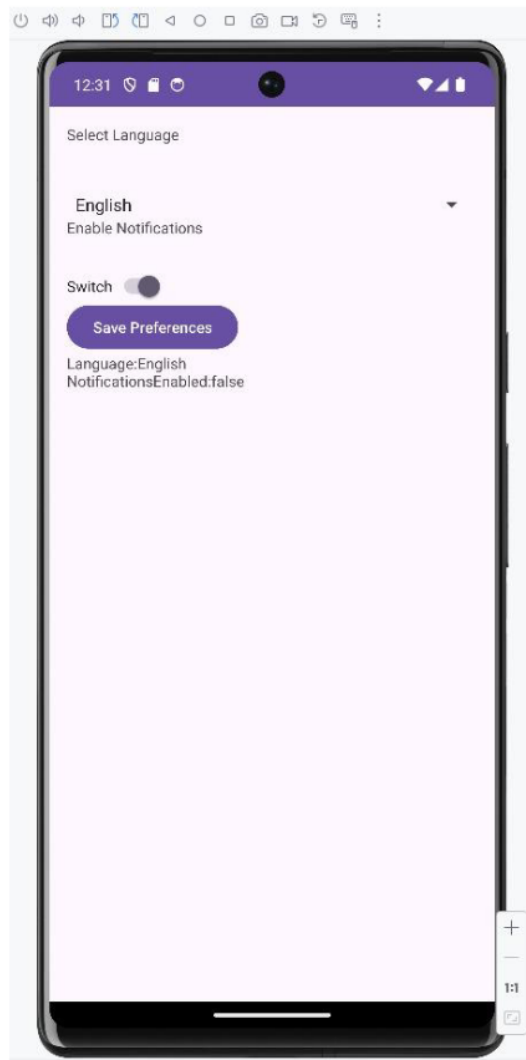
```

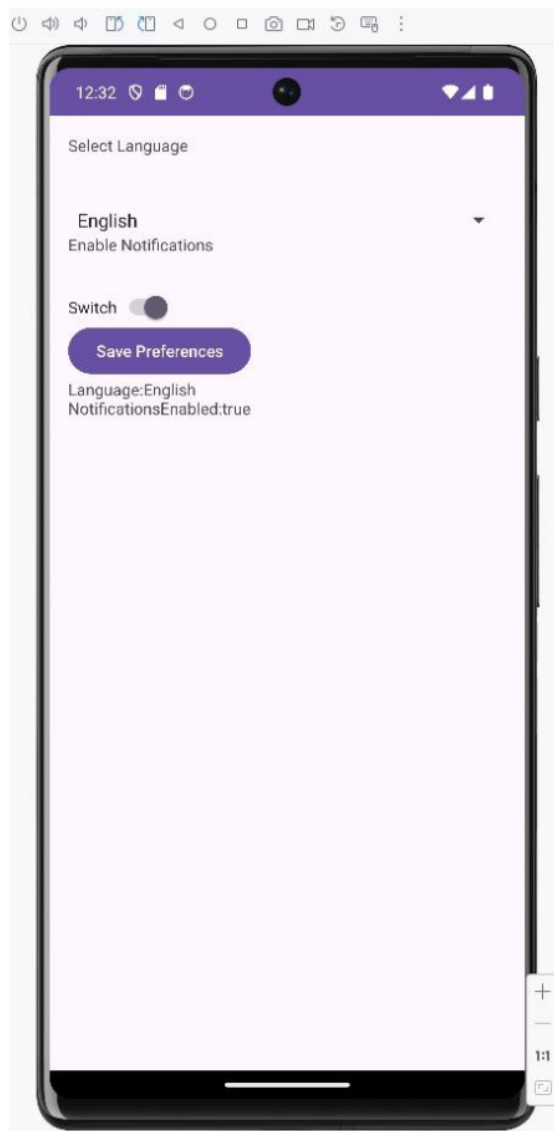
```

}
}

```

OUTPUT





Program 9: Create/Read/Write data with database(SQLite)

Design :

The image displays a mobile application UI design. On the left, a light purple form contains two text input fields labeled 'Name' and 'Email'. Below these fields is a rounded purple button labeled 'Save'. On the right, a dark teal box contains a list of UI components: 'editName', 'editEmail', 'Button', and 'TextView'.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/editName"
        android:hint="Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <EditText
        android:id="@+id/editEmail"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/buttonSave"
        android:text="Save"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
    <TextView
```



```

        android:id="@+id/textViewResult"
        android:text=""
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="20dp"/>
</LinearLayout>

```

MainActivity.java

```

package com.example.db;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    EditText editName, editEmail;
    Button buttonSave;
    TextView textViewResult;
    SQLiteHelper dbHelper;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editName = findViewById(R.id.editName);
        editEmail = findViewById(R.id.editEmail);
        buttonSave = findViewById(R.id.buttonSave);
        textViewResult = findViewById(R.id.textViewResult);
        dbHelper = new SQLiteHelper(this);
        buttonSave.setOnClickListener(v -> {
            String name = editName.getText().toString();
            String email = editEmail.getText().toString();
            dbHelper.insertUser(name, email);
            String allUsers = dbHelper.getAllUsers();
            textViewResult.setText(allUsers);
        });
    }
}

```

SQLiteHelper.java

```

package com.example.db;
import android.database.sqlite.SQLiteDatabase;
import android.content.Context;
import android.content.ContentValues;
import android.database.sqlite.SQLiteOpenHelper;
import android.database.Cursor;
public class SQLiteHelper extends SQLiteOpenHelper {
    private static final String DB_NAME = "UserDB";
    private static final int DB_VERSION = 1;
    public SQLiteHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }
    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE users (id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT, email TEXT)");
    }
    @Override

```

```

public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS users");
    onCreate(db);
}
}

public void insertUser(String name, String email) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues values = new ContentValues();
    values.put("name", name);
    values.put("email", email);
    db.insert("users", null, values);
}

public String getAllUsers() {
    SQLiteDatabase db = this.getReadableDatabase();
    Cursor cursor = db.rawQuery("SELECT * FROM users", null);
    StringBuilder sb = new StringBuilder();

    while (cursor.moveToNext()) {
        sb.append("ID: ").append(cursor.getInt(0))
        .append(", Name: ").append(cursor.getString(1))
        .append(", Email: ").append(cursor.getString(2))
        .append("\n");
    }
    cursor.close();
    return sb.toString();
}
}
}

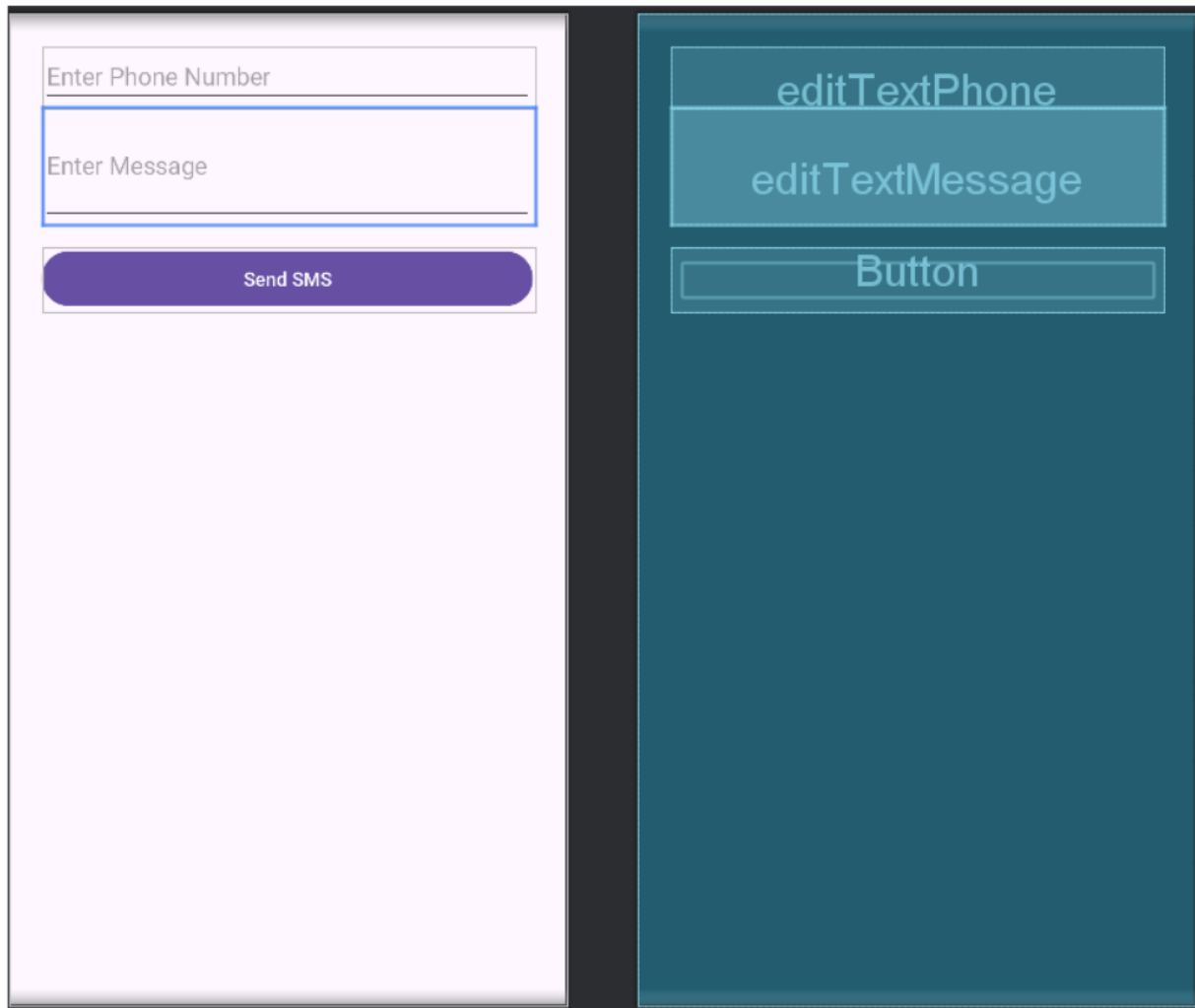
```

OUTPUT



Program 10: Create an application to send SMS and Receive SMS.

Design:



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <EditText
        android:id="@+id/editTextPhone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone Number"
        android:inputType="phone"/>

    <EditText
        android:id="@+id/editTextMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Message"
        android:inputType="textMultiLine"
    </EditText>

    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send SMS"
    </Button>

</LinearLayout>
```

```
android:minLines="3"/>
```

```
<Button
    android:id="@+id/buttonSend"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Send SMS"
    android:layout_marginTop="16dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.sms;
import android.Manifest;
import android.annotation.SuppressLint;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
public class MainActivity extends AppCompatActivity {

    EditText phoneNumberEditText, messageEditText;
    Button sendButton;
    private static final int SMS_PERMISSION_CODE = 100;
    private String permission;
    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        phoneNumberEditText = findViewById(R.id.editTextPhone);
        messageEditText = findViewById(R.id.editTextMessage);
        sendButton = findViewById(R.id.buttonSend);

        sendButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (checkPermission(Manifest.permission.SEND_SMS)) {
                    sendSMS();
                } else {
                    ActivityCompat.requestPermissions(MainActivity.this,
                        new String[]{Manifest.permission.SEND_SMS}, SMS_PERMISSION_CODE);
                }
            }
        });
    }

    private boolean checkPermission(String permission) {
```

```

        this.permission = permission;
        int check = ContextCompat.checkSelfPermission(this, permission);
        return (check == PackageManager.PERMISSION_GRANTED);
    }
    private void sendSMS() {
        String phoneNumber = phoneNumberEditText.getText().toString();
        String message = messageEditText.getText().toString();

        try {
            SmsManager smsManager = SmsManager.getDefault();
            smsManager.sendTextMessage(phoneNumber, null, message, null, null);
            Toast.makeText(this, "SMS Sent!", Toast.LENGTH_SHORT).show();
        } catch (Exception e) {
            Toast.makeText(this, "Failed to send SMS: " + e.getMessage(), Toast.LENGTH_LONG).show();
        }
    }
    @SuppressWarnings("MissingSuperCall")
    @Override

    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull
int[] grantResults) {
        if (requestCode == SMS_PERMISSION_CODE) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED)
            {
                sendSMS();
            } else {
                Toast.makeText(this, "SMS permission denied", Toast.LENGTH_SHORT).show();
            }
        }
    }
}

```

AndroidManifest.xml

```

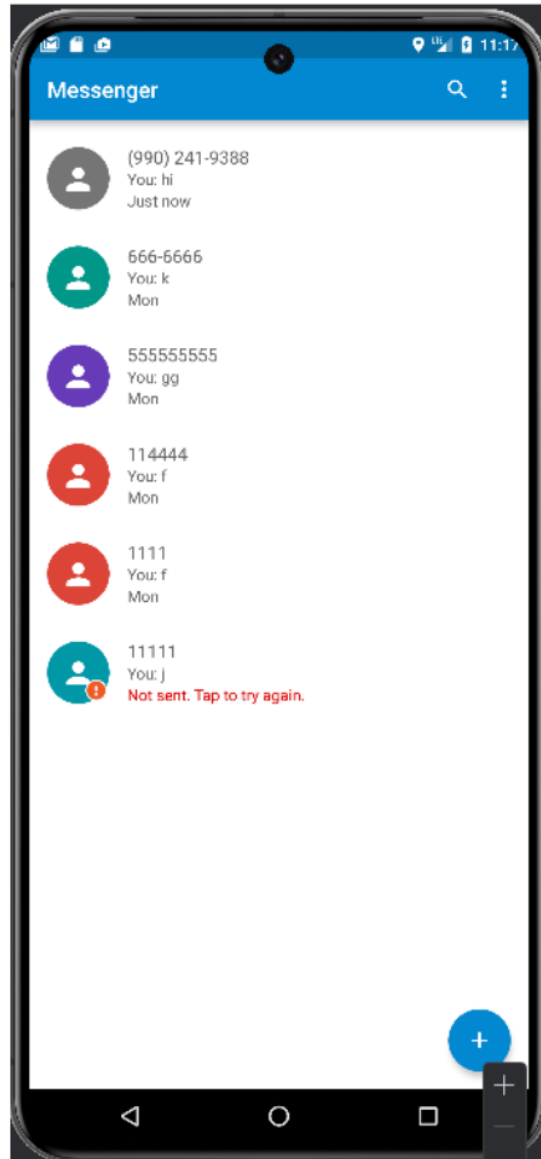
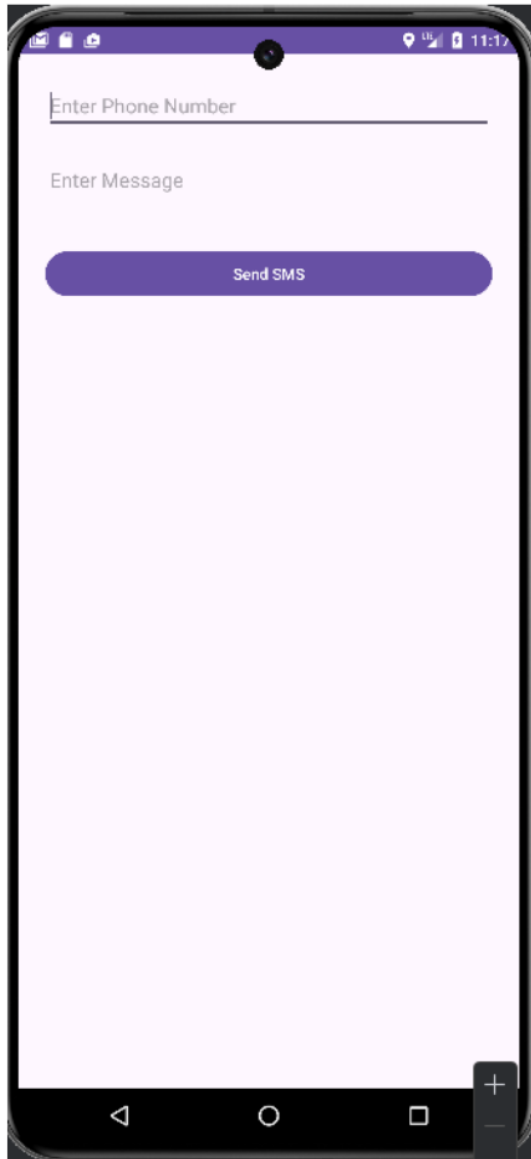
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.SEND_SMS"
        tools:ignore="PermissionImpliesUnsupportedChromeOsHardware" />
    <uses-permission android:name="android.permission.READ_PHONE_STATE"/>
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Sms"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>

```

```
<action android:name="android.intent.action.MAIN" />

<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

OUTPUT



Program 11: Create an application to send a Email.

Design

The design diagram illustrates the layout of an email application. On the left, a visual representation shows three text input fields stacked vertically, labeled 'Email Address', 'Subject', and 'Message'. Below these fields is a purple button labeled 'Send Email'. On the right, a corresponding XML layout structure is shown with labels: 'email_address' for the first input field, 'email_subject' for the second, 'email_message' for the third, and 'Button' for the 'Send Email' button.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/email_address"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Email Address"
        android:inputType="textEmailAddress" />

    <EditText
        android:id="@+id/email_subject"
```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Subject"
        android:inputType="text" />

```

```

<EditText
    android:id="@+id/email_message"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Message"
    android:inputType="textMultiLine"
    android:lines="5"
    android:gravity="top"
    android:scrollbars="vertical" />

```

```

<Button
    android:id="@+id/send_email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Send Email" />

```

```
</LinearLayout>
```

MainActivity.java

```

package com.example.email;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.example.email.R;
public class MainActivity extends AppCompatActivity {

```

```

    private EditText emailAddressEditText;
    private EditText emailSubjectEditText;
    private EditText emailMessageEditText;
    private Button sendEmailButton;

```

```

    @Override
    protected void onCreate(Bundle savedInstanceState) {

```



```

super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
emailAddressEditText = findViewById(R.id.email_address);
emailSubjectEditText = findViewById(R.id.email_subject);
emailMessageEditText = findViewById(R.id.email_message);
sendEmailButton = findViewById(R.id.send_email);
sendEmailButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        sendEmail();
    }
});
}

private void sendEmail() {
    String emailAddress = emailAddressEditText.getText().toString();
    String subject = emailSubjectEditText.getText().toString();
    String message = emailMessageEditText.getText().toString();
    Intent emailIntent = new Intent(Intent.ACTION_SEND);
    emailIntent.setType("message/rfc822");
    emailIntent.putExtra(Intent.EXTRA_EMAIL, new String[] {emailAddress});
    emailIntent.putExtra(Intent.EXTRA_SUBJECT, subject);
    emailIntent.putExtra(Intent.EXTRA_TEXT, message);

    if (emailIntent.resolveActivity(getPackageManager()) != null) {
        startActivity(Intent.createChooser(emailIntent, "Send Email Using:"));
    } else {
        Toast.makeText(this, "Email Client Not Installed", Toast.LENGTH_SHORT).show();
    }
}
}
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.SEND_SMS"
        tools:ignore="PermissionImpliesUnsupportedChromeOsHardware" />
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"

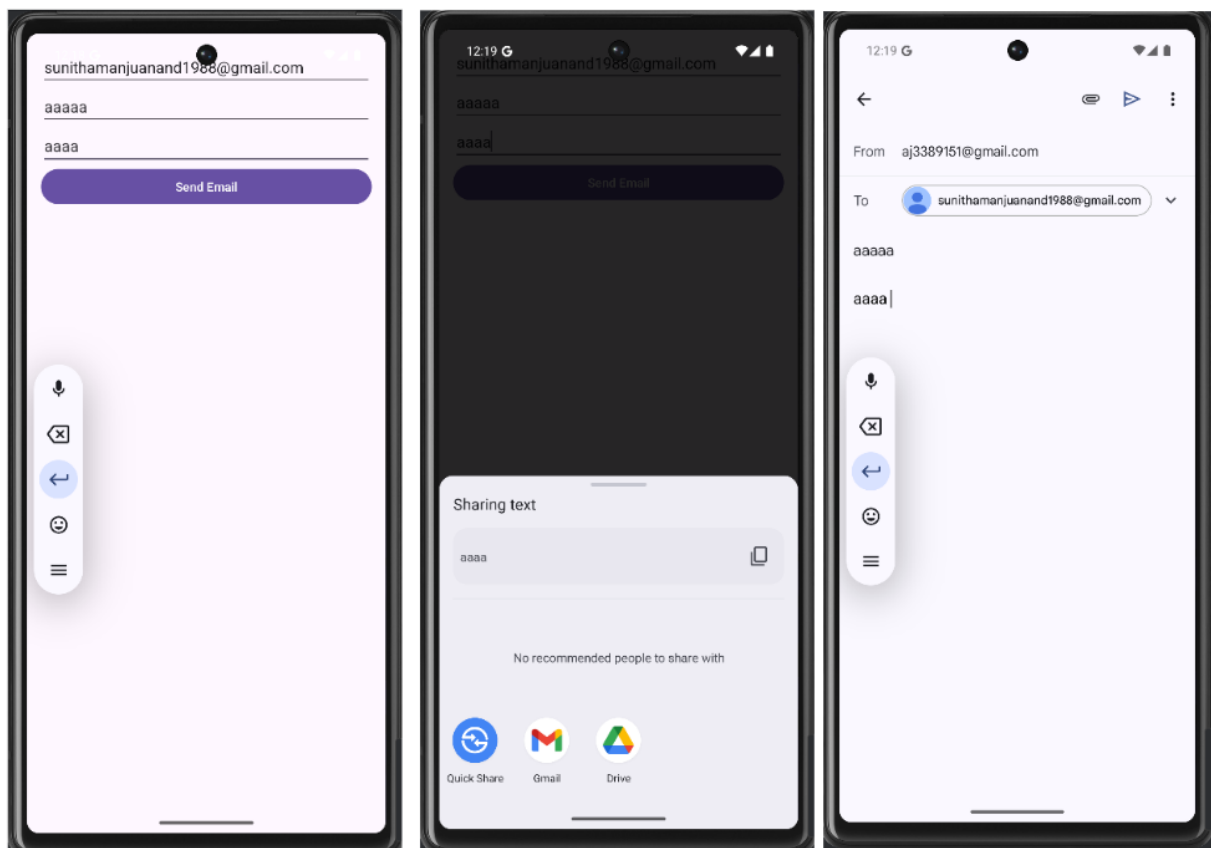
```

```

    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.Email"
    tools:targetApi="31">
    <activity
        android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
</manifest>

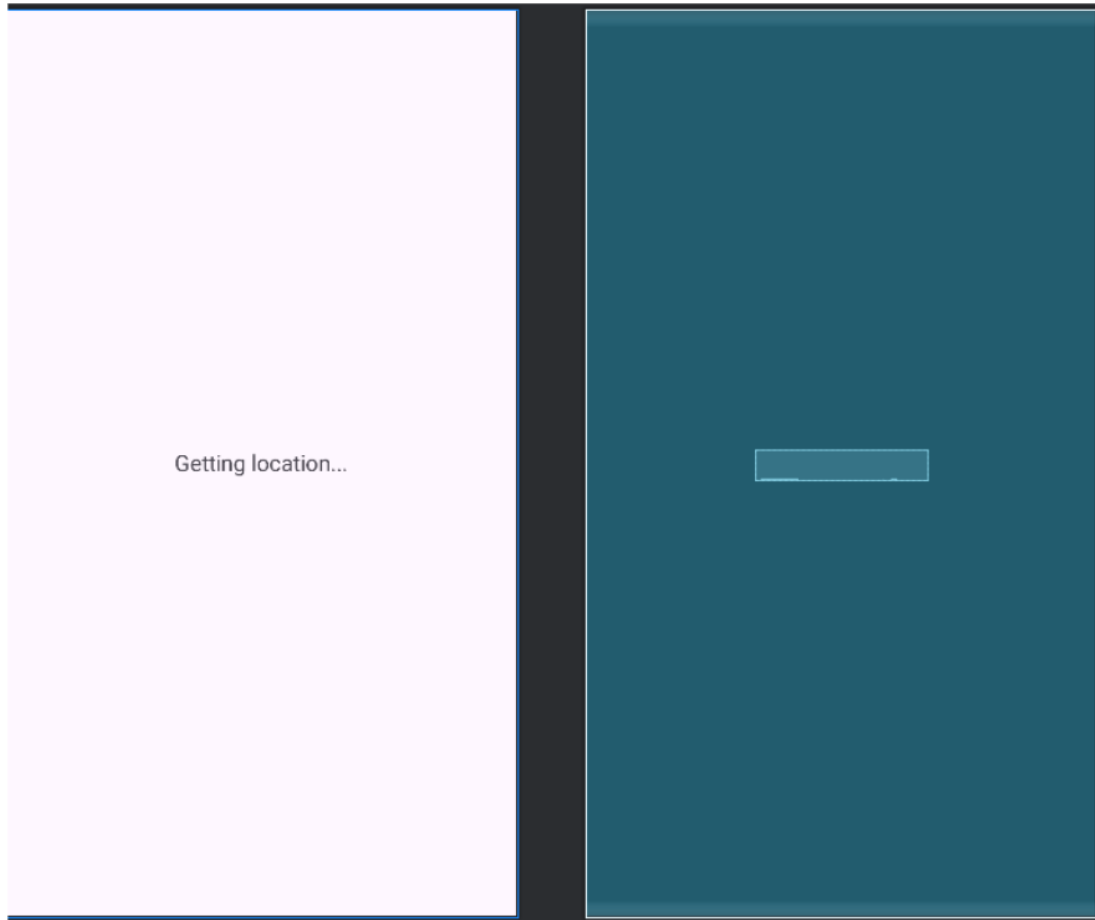
```

OUTPUT



Program 12: Display Map based on the Current/given location.

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:id="@+id/location_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Getting location..."
        android:textSize="18sp" />
</LinearLayout>
```

MainActivity.java

```
package com.example.loc;
import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;
```

```

import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
public class MainActivity extends AppCompatActivity {
    private static final int LOCATION_PERMISSION_CODE = 1;
    private LocationManager locationManager;
    private LocationListener locationListener;
    private TextView locationText;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        locationText = findViewById(R.id.location_text);
        locationManager = (LocationManager) getSystemService(LOCATION_SERVICE);
        locationListener = new LocationListener() {
            @Override
            public void onLocationChanged(@NonNull Location location) {
                double lat = location.getLatitude();
                double lon = location.getLongitude();
                locationText.setText("Latitude: " + lat + "\nLongitude: " + lon);
            }
        };
        // Check permissions
        if (ActivityCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION)
            != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this,
                new String[]{Manifest.permission.ACCESS_FINE_LOCATION},
LOCATION_PERMISSION_CODE);
        } else {
            startLocationUpdates();
        }
    }
    private void startLocationUpdates() {
        if (ActivityCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION)
            == PackageManager.PERMISSION_GRANTED) {
            locationManager.requestLocationUpdates(
                LocationManager.GPS_PROVIDER,
                5000, // Time interval (ms)
                5, // Distance (meters)
                locationListener);
        }
    }
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull
int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);

        if (requestCode == LOCATION_PERMISSION_CODE) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED)
{
                startLocationUpdates();
            }
        }
    }
}

```

```

    } else {
        Toast.makeText(this, "Location permission denied.", Toast.LENGTH_SHORT).show();
    }
}
}
}
}

```

AndroidManifest.xml

```

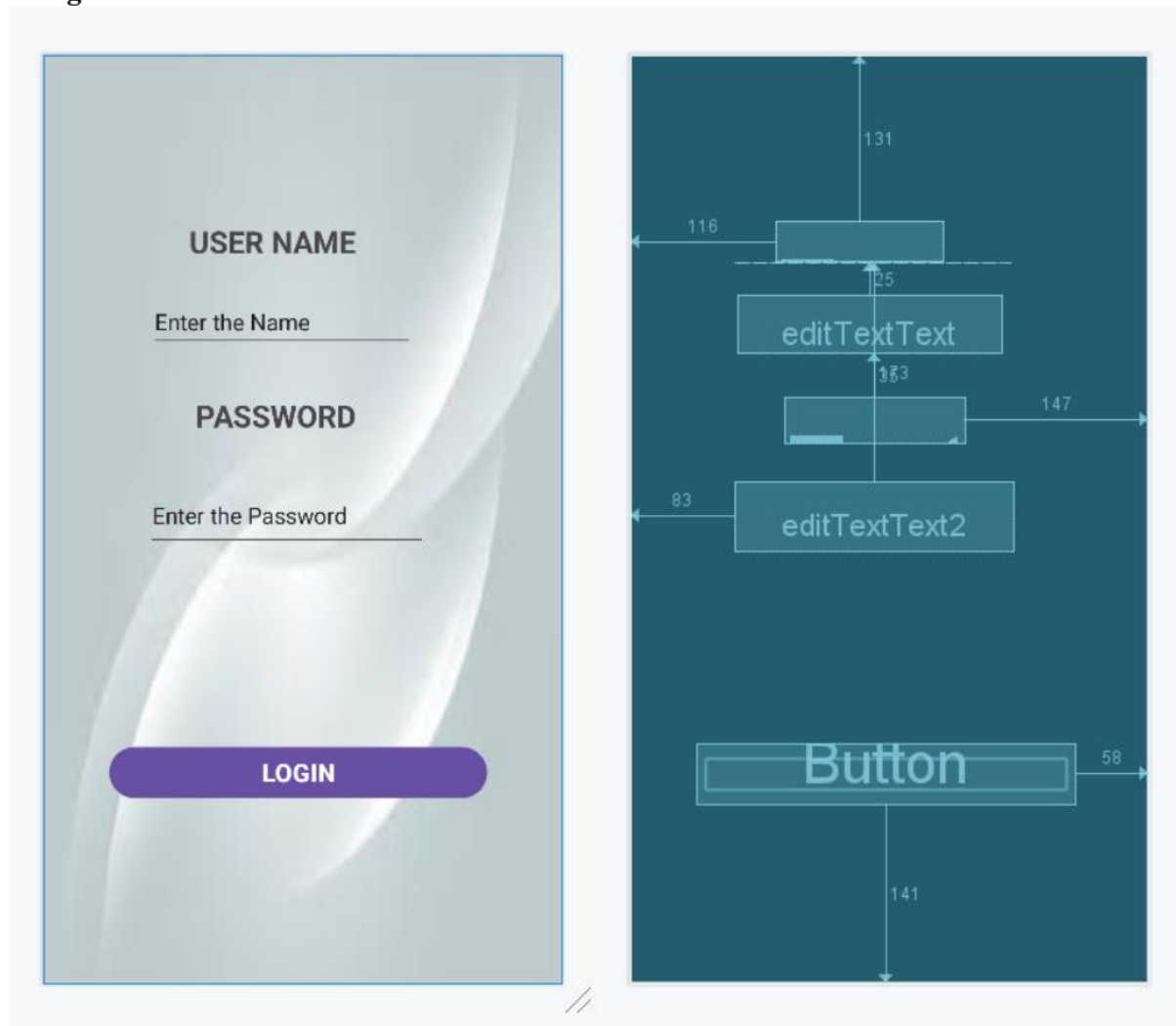
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <uses-permission android:name="android.permission.INTERNET"/>
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Loc"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>

```

OUTPUT

Program 13: Create a sample application with login module (check user name and password) On successful login change Textview “Login Successful”. On login fail alert using Toast “login fail”

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/login1"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_marginStart="116dp"
        android:layout_marginTop="131dp"
        android:text="USER NAME"
```

```

        android:textSize="24sp"
        android:textStyle="bold" />

<EditText
    android:id="@+id/editTextText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/textView"
    android:layout_marginStart="85dp"
    android:layout_marginTop="25dp"
    android:ems="10"
    android:hint="username"
    android:inputType="text"
    android:text="Enter the Name" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="143dp"
    android:layout_height="36dp"
    android:layout_below="@id/editTextText"
    android:layout_alignParentEnd="true"
    android:layout_marginTop="35dp"
    android:layout_marginEnd="147dp"
    android:text="PASSWORD"
    android:textSize="24sp"
    android:textStyle="bold" />

<EditText
    android:id="@+id/editTextText2"
    android:layout_width="222dp"
    android:layout_height="55dp"
    android:layout_below="@id/textView"
    android:layout_alignParentStart="true"
    android:layout_marginStart="83dp"
    android:layout_marginTop="173dp"
    android:ems="10"
    android:hint="password"
    android:inputType="text"
    android:text="Enter the Password" />

<Button
    android:id="@+id/button"
    android:layout_width="301dp"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="58dp"
    android:layout_marginBottom="141dp"
    android:text="LOGIN"
    android:textSize="20sp"
    android:textStyle="bold" />
</RelativeLayout>

```

MainActivity.java

```

package com.example.loginapp;
import android.os.Bundle;
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

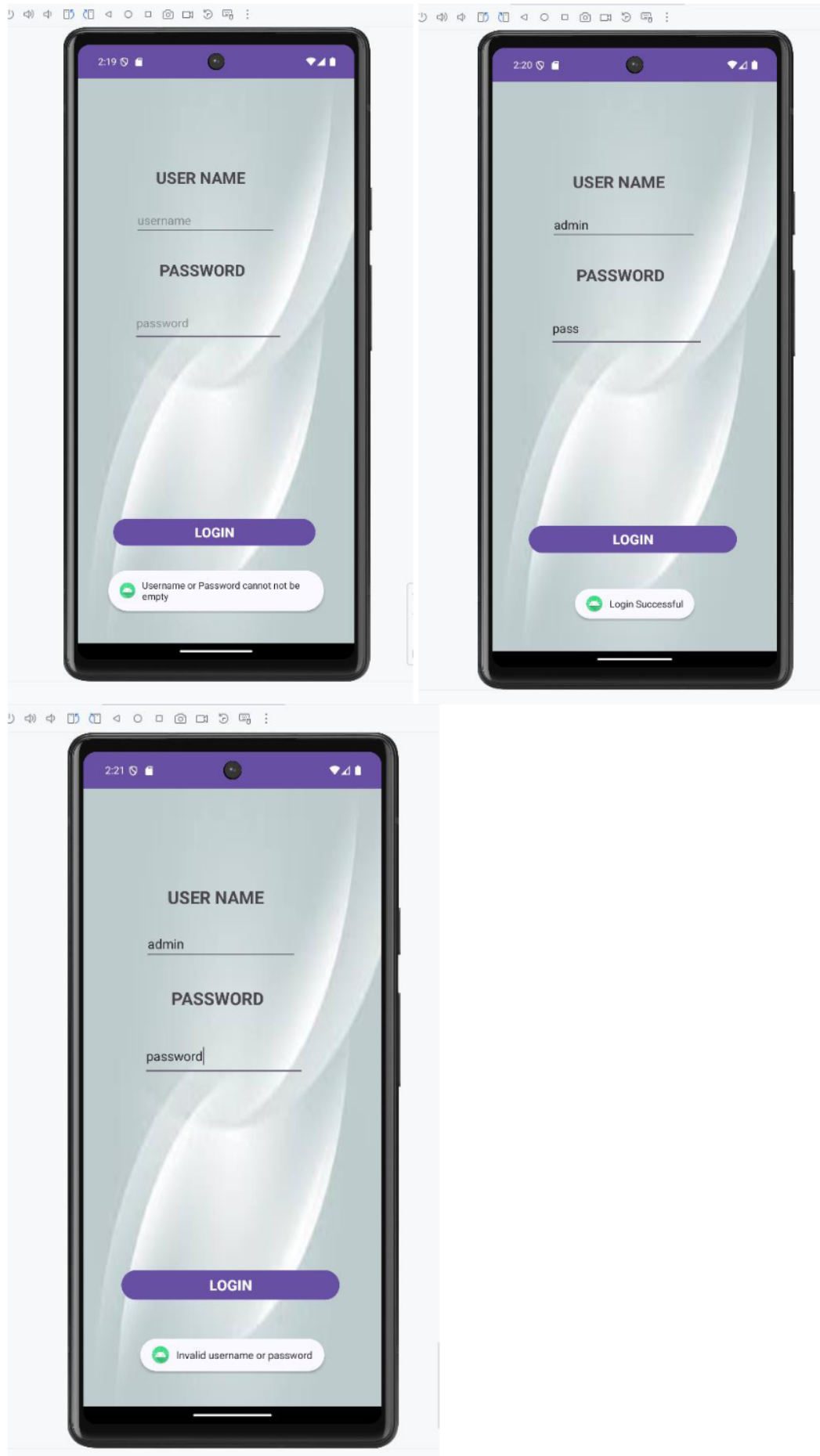
```



```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

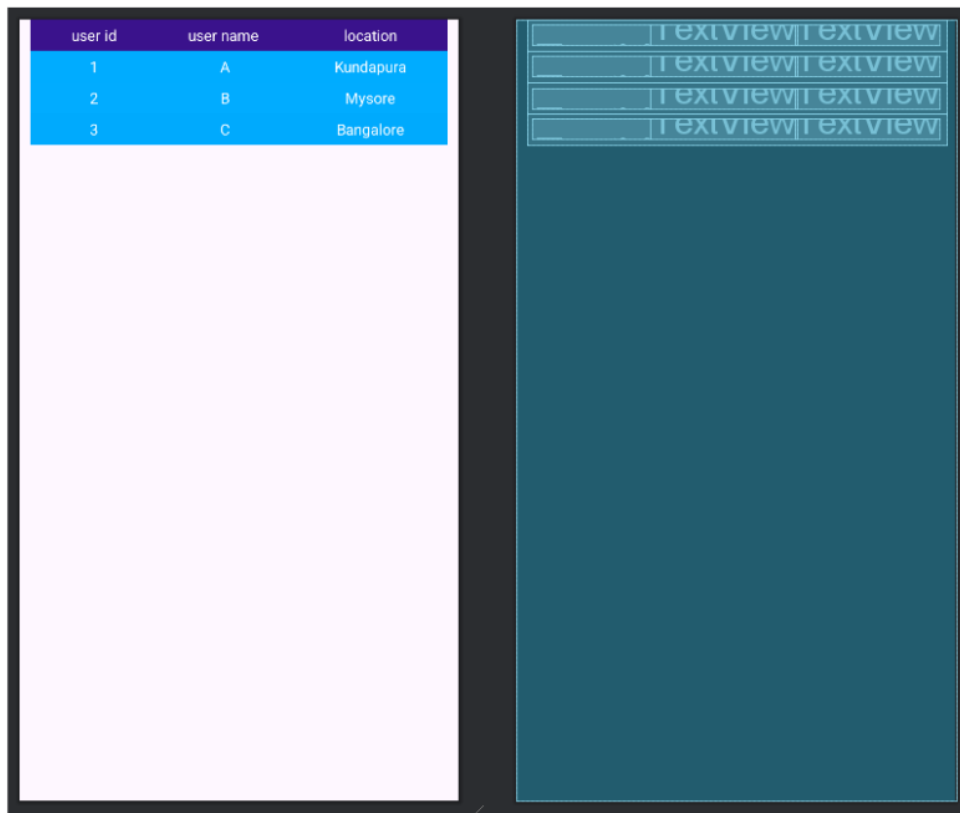
public class MainActivity extends AppCompatActivity {
    private EditText editTextUsername, editTextPassword; private Button buttonlogin;
    @Override
    protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editTextUsername=findViewById(R.id.editTextText);
        editTextPassword=findViewById(R.id.editTextText2);
        buttonlogin=findViewById(R.id.button);

        buttonlogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v)
            {
                String username=editTextUsername.getText().toString().trim();
                String password=editTextPassword.getText().toString().trim();
                if(TextUtils.isEmpty(username)||TextUtils.isEmpty(password))
                {
                    Toast.makeText(MainActivity.this," Username or Password
cannot not be empty",Toast.LENGTH_SHORT).show();
                }
                else
                {
                    if (username.equals("admin")&& password.equals("pass"))
                    {
                        Toast.makeText(MainActivity.this, "Login Successful", Toast.LENGTH_SHORT).show();
                    } else
                    {
                        Toast.makeText(MainActivity.this,"Invalid username or password",
Toast.LENGTH_SHORT).show();
                    }
                }
            }
        });
    }
}
```

OUTPUT

Program 14: Create an application to demonstrate table layout.

Design:



Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingLeft="10dp"
    android:paddingRight="10dp"
    tools:context=".MainActivity">

    <TableRow android:background="#3A128C" android:padding="5dp">
        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1" android:text="user id"
            android:textColor="@color/white"
            android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
```

```

        android:layout_weight="1" android:text="user
        name " android:textColor="@color/white"
        android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" android:text="location"
        android:textColor="@color/white"
        android:textAlignment="center"/>
    </TableRow>

    <TableRow android:background="#00ACFF" android:padding="5dp">
        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1" android:text="1"
            android:textColor="@color/white"
            android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1" android:text="A"
            android:textColor="@color/white"
            android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1" android:text="Kundapura"
            android:textColor="@color/white"
            android:textAlignment="center"/>
    </TableRow>

    <TableRow android:background="#01ACFF" android:padding="5dp">
        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="2"
            android:textColor="@color/white"
            android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1" android:text="B"
            android:textColor="@color/white"
            android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1" android:text="Mysore"
            android:textColor="@color/white"
            android:textAlignment="center"/>
    </TableRow>

    <TableRow android:background="#00AAFD" android:padding="5dp">

```

```
<TextView android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1" android:text="3"
    android:textColor="@color/white"
    android:textAlignment="center"/>

<TextView android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1" android:text="C"
    android:textColor="@color/white"
    android:textAlignment="center"/>

<TextView android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1" android:text="Bangalore"
    android:textColor="@color/white"
    android:textAlignment="center"/>
</TableRow>
</TableLayout>
```

MainActivity.java

```
package com.example.demo11;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    });
}
```

Output:A smartphone screen displaying a table with three columns: 'user id', 'user name', and 'location'. The table has three rows of data. The first row shows '1', 'A', and 'Kundapura'. The second row shows '2', 'B', and 'Mysore'. The third row shows '3', 'C', and 'Bangalore'. The table is set against a light purple background. The smartphone's status bar at the top shows the time as 4:06 and various icons for signal, Wi-Fi, and battery.

user id	user name	location
1	A	Kundapura
2	B	Mysore
3	C	Bangalore

